

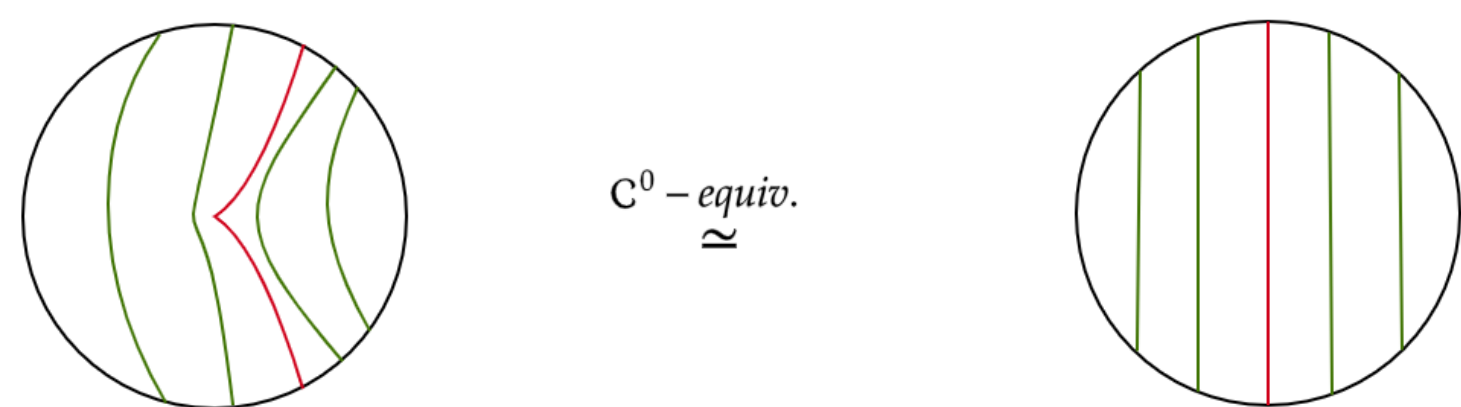
SINGULARITIES IN REAL GEOMETRY

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Motivation

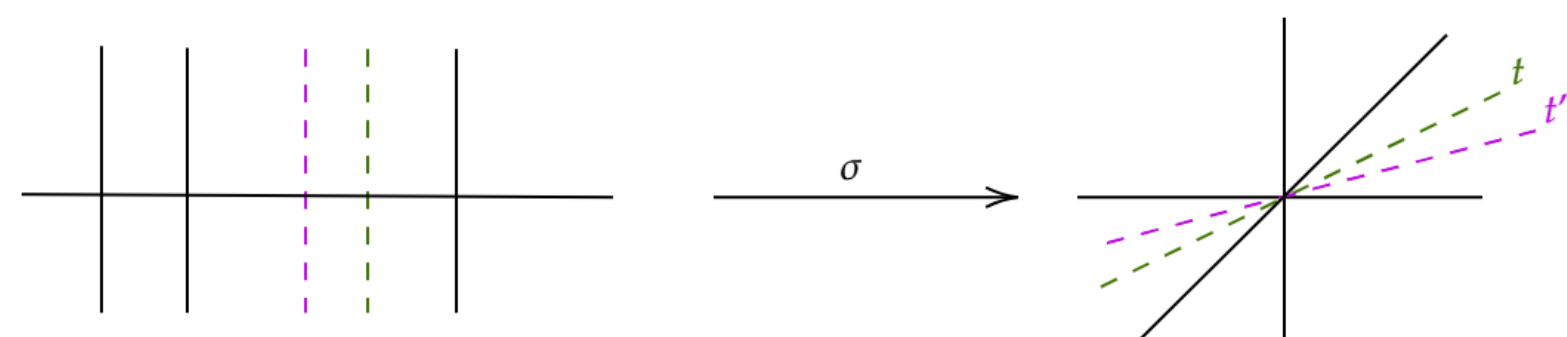
The study and classification of singularities in real analytic geometry requires the choice of a relevant equivalence relation. We say that two real analytic function germs $f, g : (\mathbb{R}^d, 0) \rightarrow (\mathbb{R}, 0)$ are $\mathcal{C}^k$ -equivalent if there exists $\varphi : (\mathbb{R}^d, 0) \rightarrow (\mathbb{R}^d, 0)$ a $\mathcal{C}^k$ -diffeomorphism such that $f = g \circ \varphi$. The topological equivalence is too flexible because it identifies for example all germs $(x, y) \rightarrow x^2 - y^{2k+1}$ with the non-singular germ $(x, y) \rightarrow x$.



On the other hand, the $\mathcal{C}^1$ -equivalence is already too coarse as shown by the example below.

Example : The Whitney family

For $t \in]0, 1[$, we consider $f_t(x, y) = xy(y - x)(y - tx)$. Then f_t and $f_{t'}$ are $\mathcal{C}^1$ -equivalent if and only if $t = t'$. However, Kuo make the following observation in [6]. Denote $\sigma : X \rightarrow \mathbb{R}^2$ the blow-up of the plane in the origin.



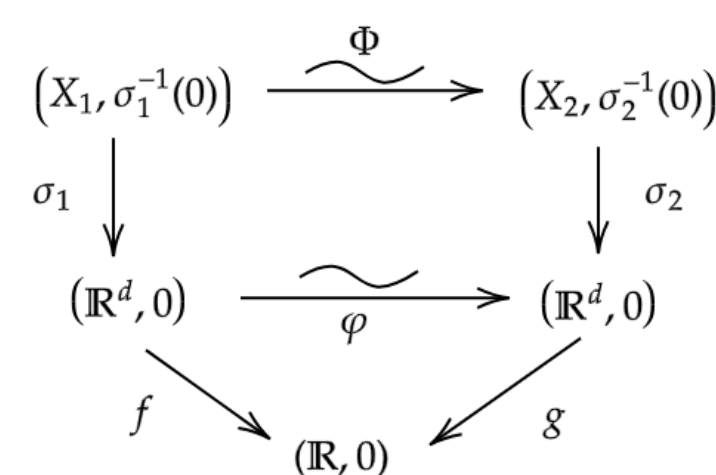
Then the germs $f_t \circ \sigma : X \rightarrow \mathbb{R}$ are all analytically equivalent.

The blow-analytic equivalence

The blow-analytic equivalence generalizes the previous situation.

Definition

We say that two real analytic function germs $f, g : (\mathbb{R}^d, 0) \rightarrow (\mathbb{R}, 0)$ are blow-analytic equivalent if there exist an homeomorphism $\varphi : (\mathbb{R}^d, 0) \rightarrow (\mathbb{R}^d, 0)$, two real modifications $\sigma_i : (X_i, \sigma_i^{-1}(0)) \rightarrow (\mathbb{R}^d, 0)$ and a real analytic isomorphism $\Phi : (X_1, \sigma_1^{-1}(0)) \rightarrow (X_2, \sigma_2^{-1}(0))$ such that the following diagram commutes :

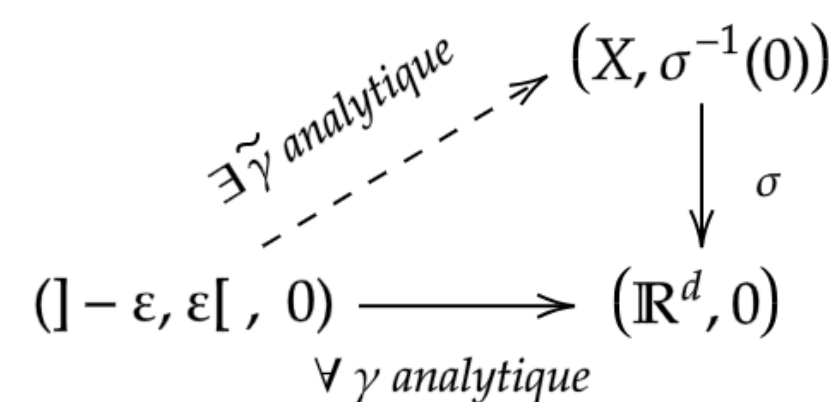


Theorem

A family of isolated singularities has only a finite number of blow-analytic classes i.e let $F : (\mathbb{R}^d \times I, \{0\} \times I) \rightarrow (\mathbb{R}, 0)$ be a real analytic function germ such that for every $t \in I$, $f_t(x) = F(x, t) : (\mathbb{R}^d, 0) \rightarrow (\mathbb{R}, 0)$ has an isolated singularity at the origin. Then the germs $(f_t)_{t \in I}$ define a finite number of blow-analytic classes.

Seeking for invariants

In [4], Fukui points out that every real modification $\sigma : (X, \sigma^{-1}(0)) \rightarrow (\mathbb{R}^d, 0)$ has the arc lifting property summarized by the following diagram :



Thus he defines the following set

$$A(f) = \{ \text{ord}_t(f \circ \gamma) \mid \gamma : (] - \varepsilon, \varepsilon[, 0) \rightarrow (\mathbb{R}^d, 0) \text{ analytic} \} \subset \mathbb{N}^* \cup \{\infty\}$$

which is an invariant of the blow-analytic equivalence.

Properties of the Fukui invariant

- 1) If $k \in A(f)$, then $\forall l \in \mathbb{N}^*, kl \in A(f)$. In particular $A(f)$ is an infinite set.
- 2) $\min A(f) = \text{mult}_0(f)$.
- 3) f has no critical points $\iff 1 \in A(f) \iff A(f) = \mathbb{N}^* \cup \{\infty\}$.

Examples

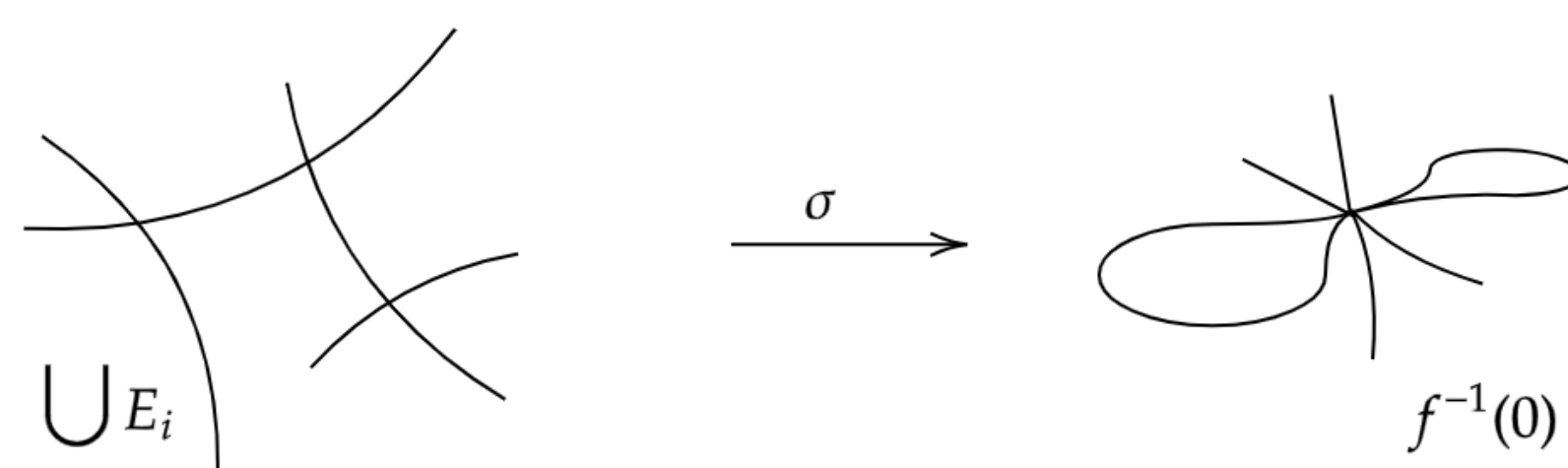
- 1) Let $f(x, y) = x^2 \pm y^2$. Then we have $A(f) = 2\mathbb{N}^* \cup \{\infty\}$.
- 2) If f is normal crossing i.e $f(x_1, \dots, x_d) = x_1^{m_1} \dots x_d^{m_d}$, then

$$A(f) = m_1\mathbb{N}^* + \dots + m_d\mathbb{N}^* \cup \{\infty\}.$$

General case

Thanks to Hironaka desingularisation theorem, there exists $\sigma : (X, \sigma^{-1}(0)) \rightarrow (\mathbb{R}^d, 0)$ a real modification such that $f \circ \sigma$ is normal crossing. Let $\bigcup_{i \in J} E_i$ be the decomposition into irreducible components of $(f \circ \sigma)^{-1}(0)$ and $D = \text{div}(f \circ \sigma) = \sum_{i \in J} m_i E_i$ the associated principal divisor. We denote $\bigsqcup_{I \subset J} E_I^0$ the canonical stratification of $(f \circ \sigma)^{-1}(0)$ into smooth subvarieties and for $I \subset J$ such that $I = \{i_1, \dots, i_p\}$ let us put

$$\Omega_I(f) = m_{i_1}\mathbb{N}^* + \dots + m_{i_p}\mathbb{N}^* \cup \{\infty\}$$

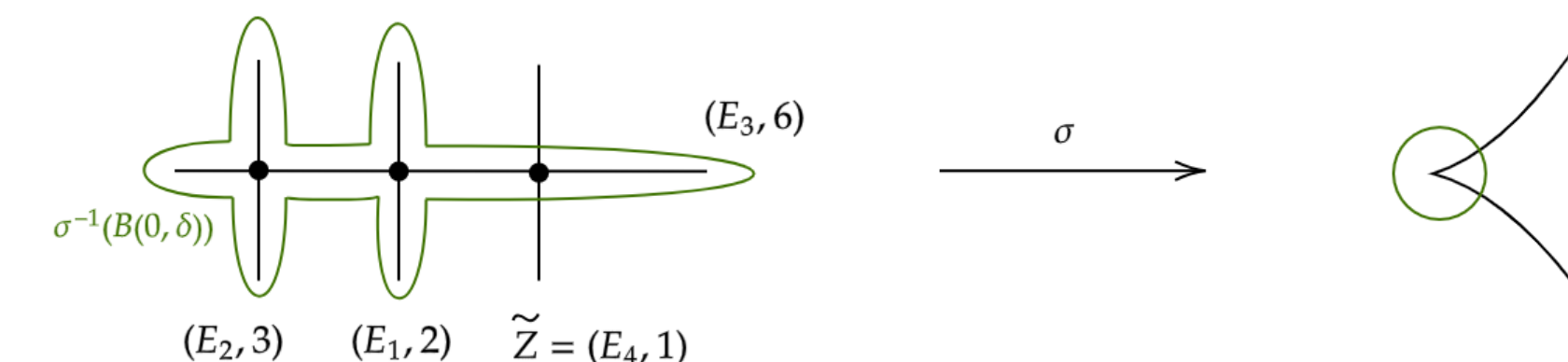


Theorem

Then $A(f) = \bigcup_{I \in C} \Omega_I(f)$ where $C = \{ I \subset J \mid \sigma^{-1}(0) \cap E_I^0 \neq \emptyset \}$ (c.f [5] Theorem I)

An example

Let $f(x, y) = y^2 - x^3$, we can make 3 punctual blow-ups to get a modification as follows :



therefore $A(f) = 2\mathbb{N}^* \cup 3\mathbb{N}^* \cup (6\mathbb{N}^* + \mathbb{N}^*) \cup \{\infty\} = \{2, 3, 4, 6, 7, 8, 9, 10, \dots\}$.

The geometry of arc spaces

The theory of motivic integration (as developed in [2]) naturally provides more subtle invariants that take into account the geometry of all arcs that will carry out a given order. For $n \geq 1$, denote $\mathcal{L}_n(\mathbb{R}^d, 0)$ the space of truncated analytic arcs starting from the origin i.e

$$\mathcal{L}_n(\mathbb{R}^d, 0) = \{ \gamma \in \mathbb{R}[[t]]/(t^{n+1}) \mid \gamma(0) = 0 \} \simeq \mathbb{R}^{nd}.$$

We then consider the space

$$\mathcal{X}_n(f) = \{ \gamma \in \mathcal{L}_n(\mathbb{R}^d, 0) \mid \text{ord}_t(f \circ \gamma) = n \}$$

which is a constructible subset of $\mathbb{R}^{nd}$.

A richer invariant

The motivic zeta function of f is a generating series with coefficients in the so called Grothendieck ring of varieties $K_0(\mathbb{R}\text{Var})$ defined by

$$Z_{\text{mot}}(f; T) = \sum_{n \geq 1} [\mathcal{X}_n(f)] \mathbb{L}^{-nd} T^n$$

Similarly, there exists a formula to express the motivic zeta function in terms of a modification $\sigma : (X, \sigma^{-1}(0)) \rightarrow (\mathbb{R}^d, 0)$ such that $f \circ \sigma$ and $\text{jac } \sigma$ are normal crossing.

A Denef-Loeser type formula

$$Z_{\text{mot}}(f; T) = \sum_{I \subset J} (\mathbb{L} - 1)^{|I|} [E_I^0 \cap \sigma^{-1}(0)] \prod_{i \in I} \frac{\mathbb{L}^{-\nu_i} T^{N_i}}{1 - \mathbb{L}^{-\nu_i} T^{N_i}}$$

where $N_i = \text{ord}_{E_i}(f \circ \sigma)$ and $\nu_i = 1 + \text{ord}_{E_i}(\text{jac } \sigma)$.

References

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